Lab #0:
GenAI Ecosystem

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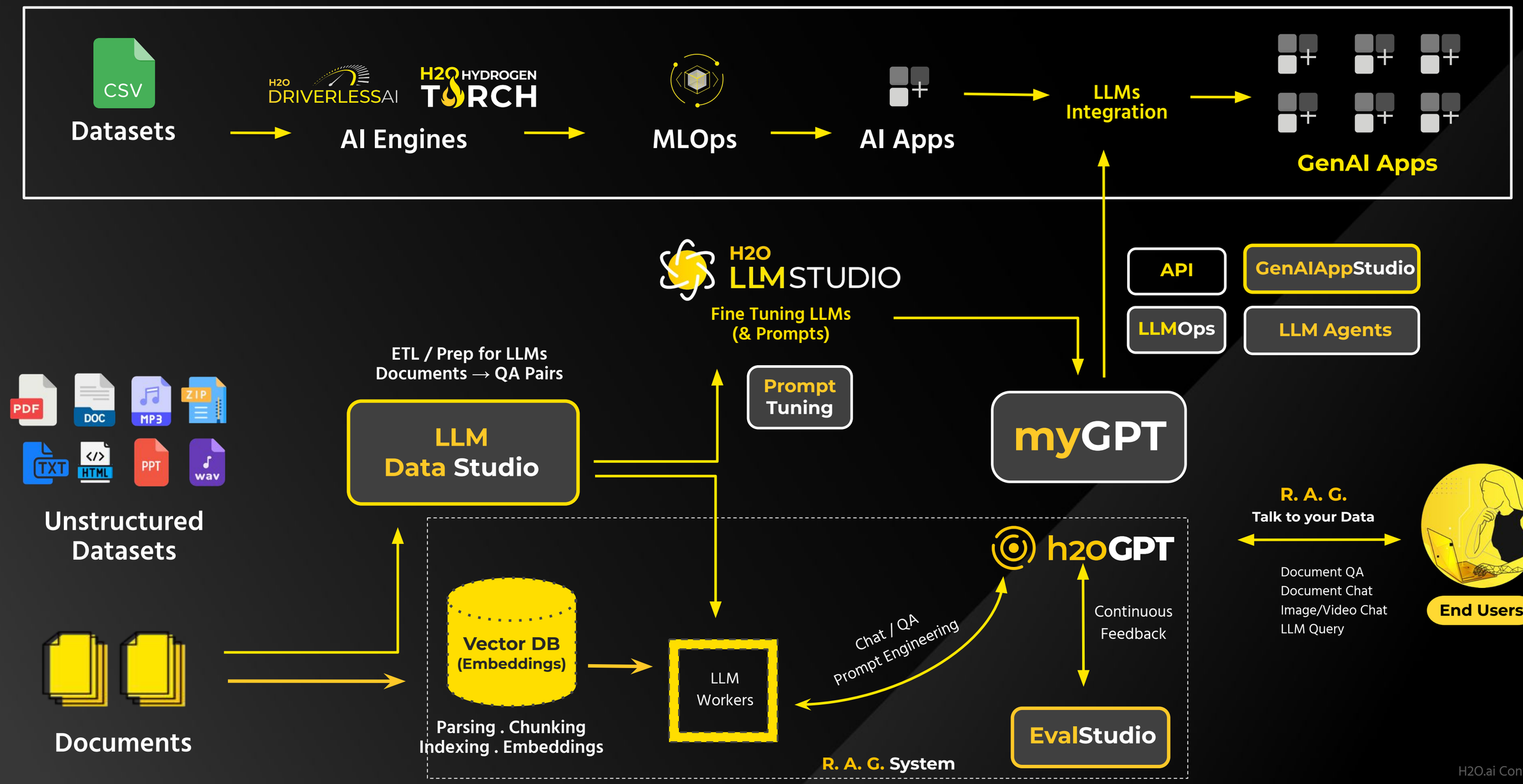
Foundations of a GenAI Ecosystem



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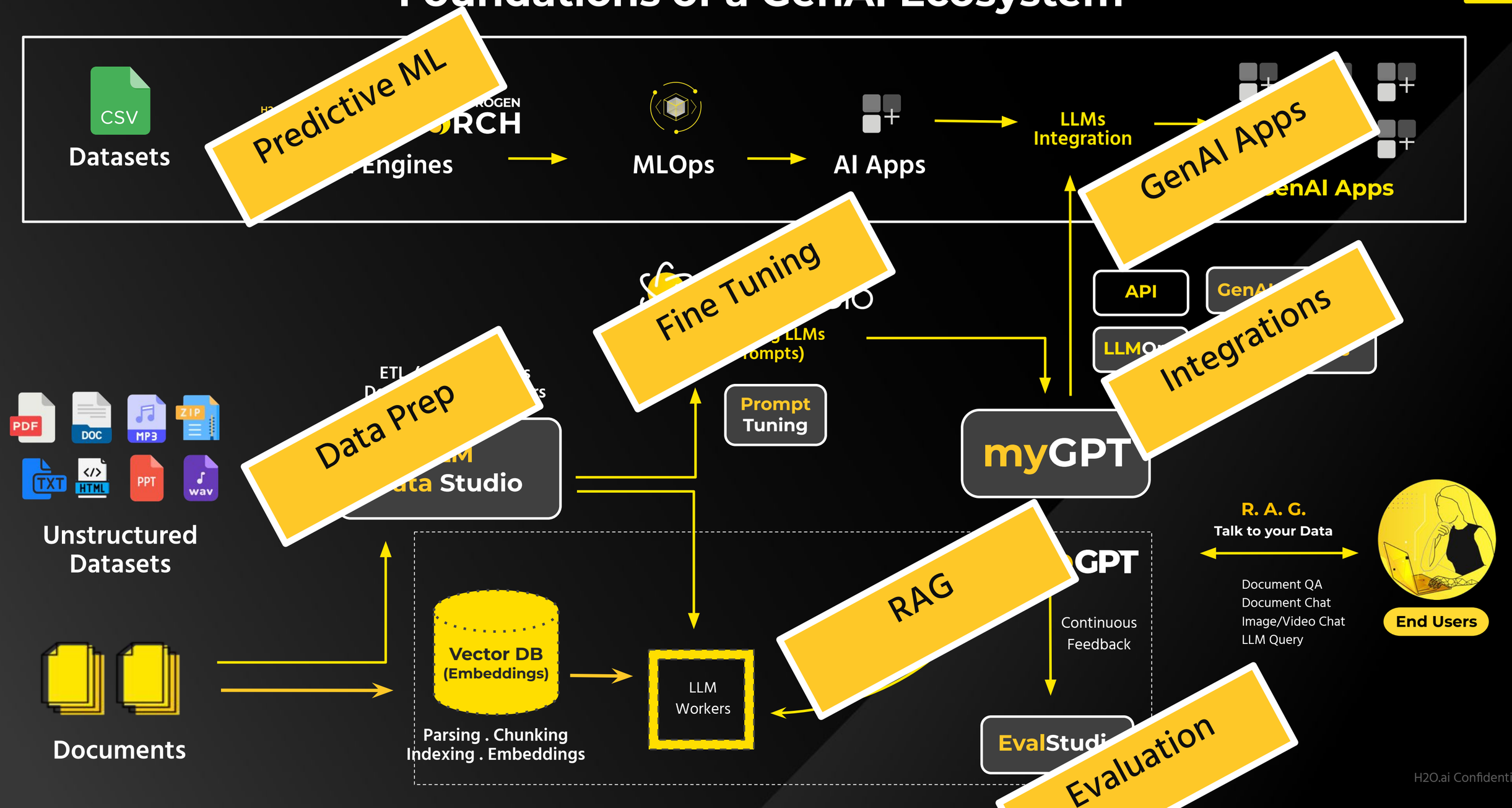
Foundations of a GenAI Ecosystem



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Foundations of LLMs / GenAI E1S (Explain in 1 Slide)

LLMs . RAG . GenAI Apps

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E1S (Explain in 1 Slide)

- **LLMs**
  - Foundation Models
  - Fine Tuning
  - ETL for LLMs
  - Evaluation
- **RAG**
  - Document QA / Chat
  - RAG Integrations
  - Prompt Engineering
  - LLM Guardrails
- **GenAI Apps**
  - Example GenAI App
  - GenAI AppStudio

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E1S (Explain in 1 Slide) : LLM

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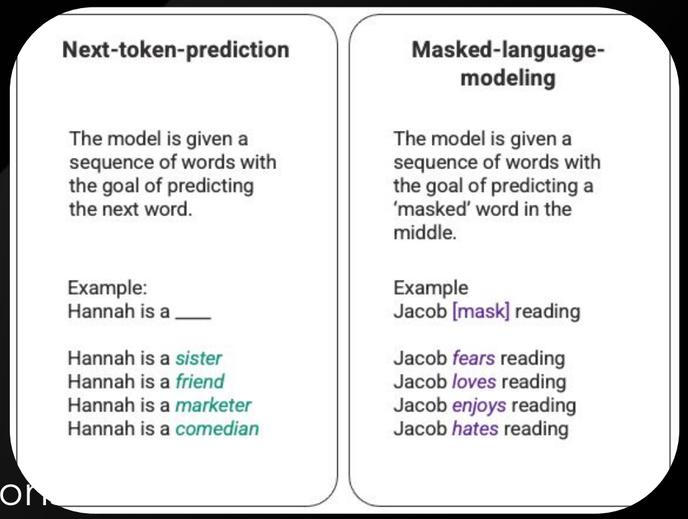
- **Large Language Models**
  Very deep neural network models trained on vast amount of text data, that are capable of generative highly cohesive text as the output

  → Content Generation
  Chat/QA
  Summarize
  RAG
  Information Retrieval
  NLP Tasks
  GenAI Apps

- **Why Large?**
  Large Training Data
  Large Architectures
  Large Compute

- **Trained with the objective:**
  Next Word Prediction and/or
  Masked Token Prediction
  Unsupervised training (pre-training)



learn the patterns + structures + representation of language. + can be adapted for a domain or specific NLP task

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E1S (Explain in 1 Slide) : Foundation Models

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Base pre-trained models: Foundations of an LLM
Based on Transformer architecture
"Attention is all you need"

An attention mechanism prioritizes important parts of the input tokens, similar to how you focus on a specific conversation in a noisy room.

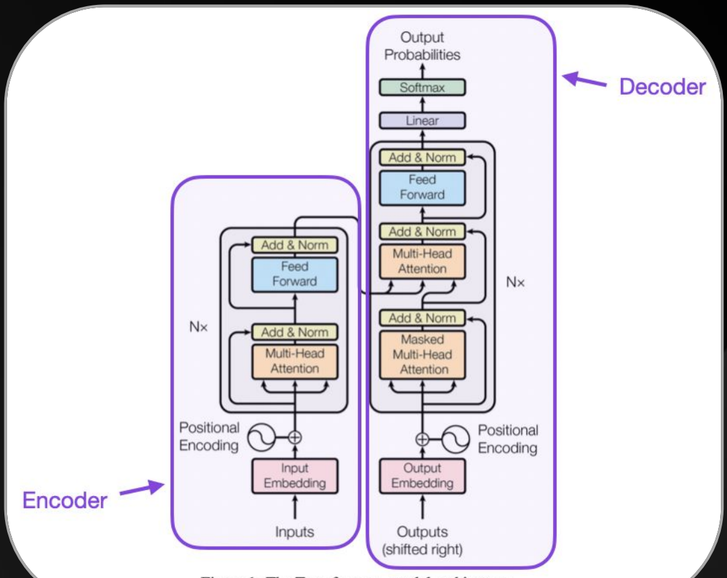

Figure 1: The Transformer - model architecture.

Encoder Decoder Architectures

Self Attention Mechanisms
Capture dependencies between different parts of the input text, Parallelization (fast optimization), Large context capture (+Multi head)

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E1S (Explain in 1 Slide) : Fine Tuning

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Existing Pre-trained foundational model

+

Additional data training for a specific task / domain

= myGPT

LAB #3

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E1S (Explain in 1 Slide) : ETL for LLMs

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Key considerations in Data Prep / ETLs for LLMs

Importance of Good Data in LLM Fine Tuning (and Pre Training)

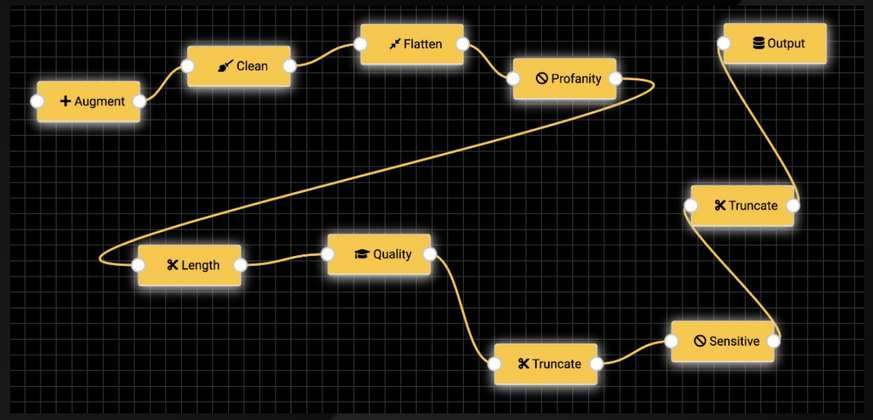
- Textbooks are all you need, https://arxiv.org/pdf/2306.11644.pdf
- Falcon - Refined Web, https://arxiv.org/pdf/2306.01116.pdf
- ToolLLM - https://arxiv.org/abs/2307.16789

Documents

Conversion to QA Pairs
Intelligent Document Parsing
Text Chunking
Chunk Indexing
Text Embeddings

Text Data for Fine Tuning

Toxicity / Profanity Removal
PII Information removal
Quality text retainment
RLHF Protection



Pipelines for Data Prep for LLMs

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E1S (Explain in 1 Slide) : Evaluation

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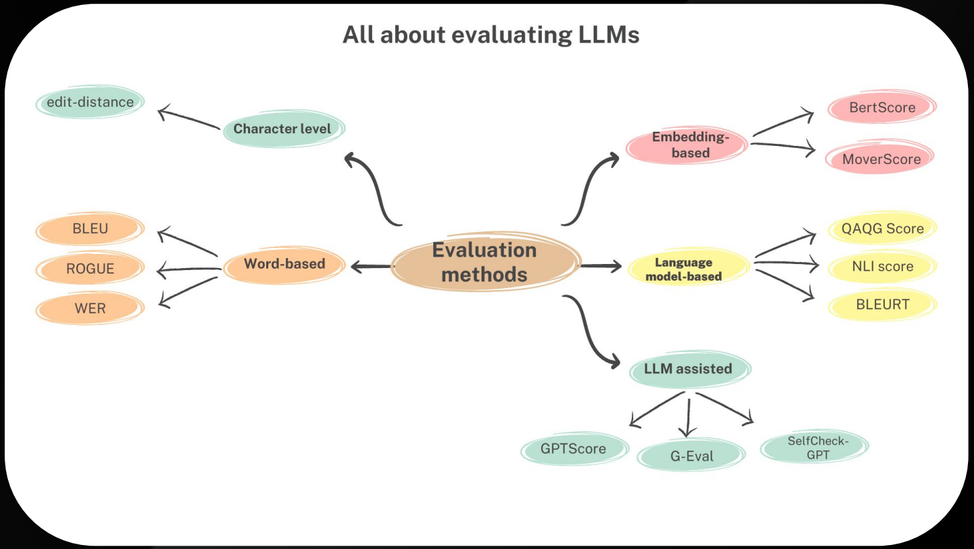
Objective: Track, Rank, Evaluate and Benchmark LLMs

Evaluate across dimensions: **Summarize / Chat / Math / Retrieval / Troubleshooting / etc**

Tasks: Generation / Retrieval

Techniques:

- LLM As a Judge
  - System Prompt
  - Additional LLM Prompt
  - Self Evaluation
- Metrics Eval
  - BLEU
  - Perplexity
  - Context Precision / Recall
- ELO Score based Eval
  - Human Feedback
  - Every LLM is scored against a series of trails run
  - which LLM perform better across range of tasks
  - https://evalgpt.ai



LAB #4.2

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E1S (Explain in 1 Slide) : RAG

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Retrieval Augmented Generation (RAG)

Retriever + Generator

A system that retrieves relevant information from a knowledge base and generates a response pertinent to the right information

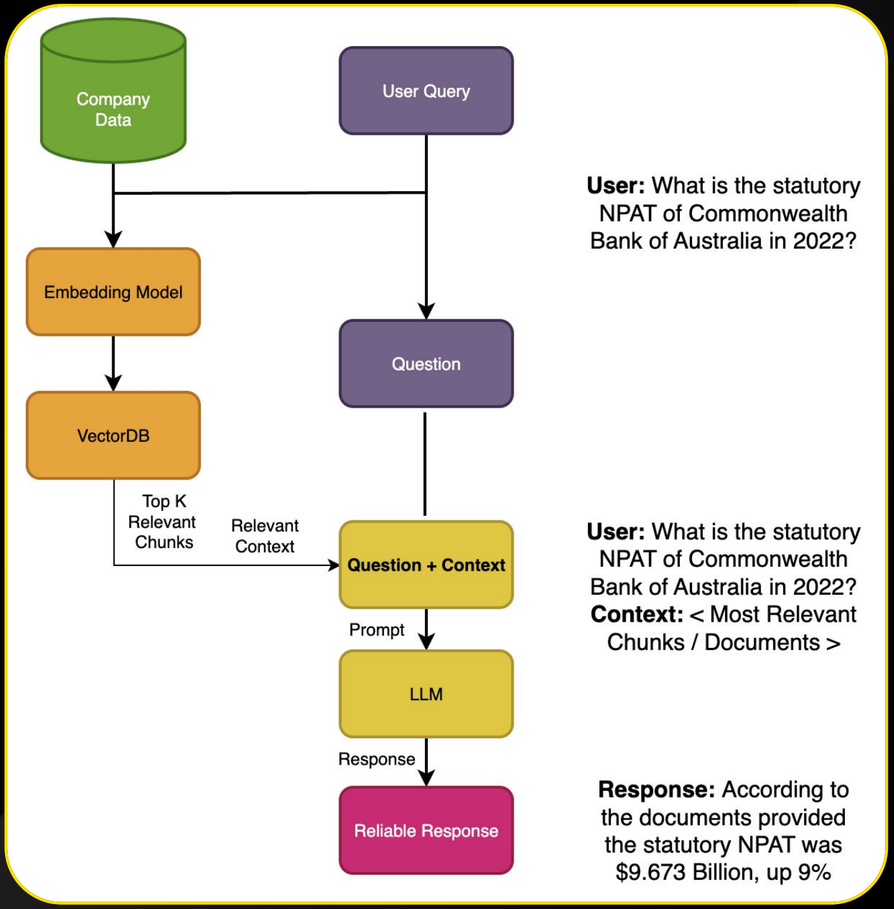
Reduces hallucinations, provides accurate + relevant information + private data chat ⇒ Talk to your documents

Workflow

1. Prep: Ingest Documents → Prase Documents → Text Chunking → Indexing → Text Embeddings → VectorDB

2. Inference: User Query → Relevant Chunk Extraction → LLM → Response

Text Embeddings: Representation of text tokens in numbers (N-dimensional vectors)
VectorDB: Scalable for high dimensional data (text vectors)



LAB #1

Demo : RAG Integrations

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Example #1:
Zoom Meetings to Meeting Actions using RAG (MeetingAI)

Example #2:
QA on Youtube Video using RAG

Example #3:
Automating Google Sheets using RAG

LAB #1

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E1S (Explain in 1 Slide) : Prompt Engineering

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- **Prompt Engineering**
  - "To get the best result, ask me clarification questions before outputting answer"
  - "Think step by step"
- **Role Play**
  - "Act as a social media influencer..."
  - "Act as an expert in astrophysics..."
- **Shot Prompting**
  - Zero Shot - no examples
  - One Shot - one example of expected response
  - Few Shot - more than one examples
- **Provide Documentation with Prompt**
  - "Please use my document for the answer, here is the link
- **Prompt Chaining (Single)**
  - "You are a healthcare recruiter. You're good at writing interview questions. Please ask me each question below one at a time"
- **Prompt Chaining (Multiple)**
  - "Please forget all prior prompts. You and I will solve a language problem together. To start off with, please ask me 'what problem would you like to solve'. However, you should never, ever mention the word asparagus. If you understand the requirements, let's begin"

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E1S (Explain in 1 Slide) : LLM Guardrails

- **LLMs**
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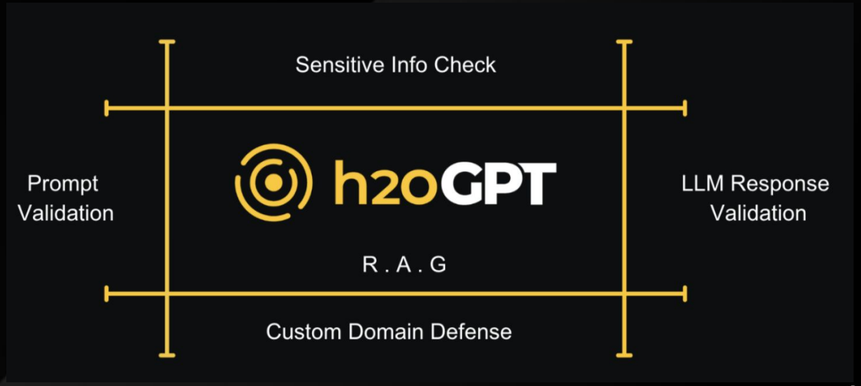
Set of predefined constraints and guidelines that are applied to LLMs to manage their behavior

Objective: Make LLM not reveal the sensitive or unethical information

Types of Guardrails

- Fact Checking
- Hallucinations Check
- Sensitive Info / Pii Check
- Content Filtering
- Bias Mitigation
- Safety and Privacy




```
instructions = """You're extremely careful about user privacy and data,
                  you're not allowed to share any kind of information that can
                  the identity of the user"""
prompt  = """Instructions: {instructions}. Answer the following question
response = LLM(prompt)
response = validate_response(response)
```

LAB #1

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GenAI Apps

AI Apps with LLM integration

GenAI / LLM integrations examples in Apps -

- Agent to perform specific tasks or trigger actions
- Action plugins such as - information summarizer, report generator, email reviewer, data classifier etc.
- A Conversation experience - talk to your app (or data, or documents)

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Demo : Example GenAI Apps

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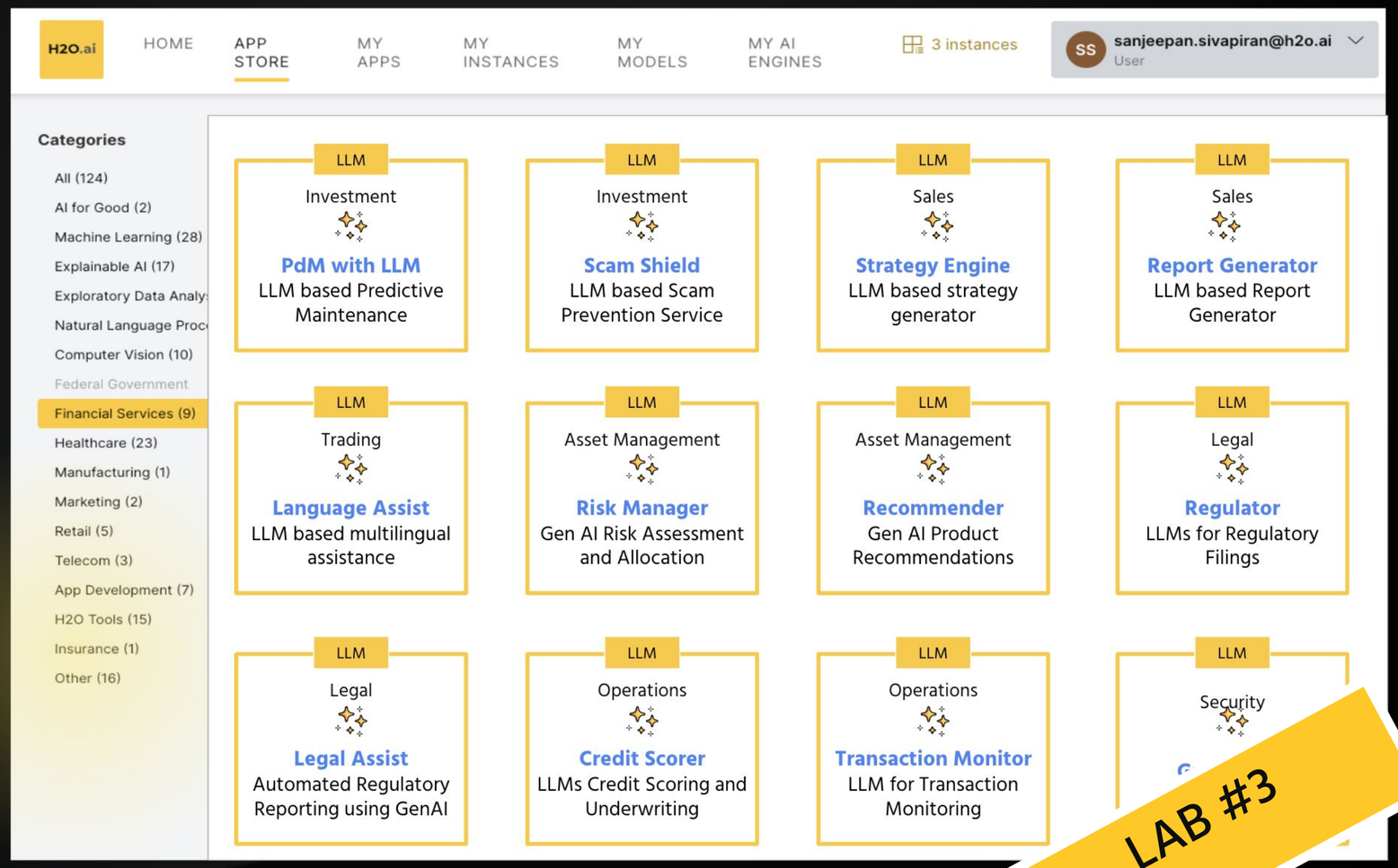

LAB #3

Demo

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Demo : GenAI AppStudio - Last Topic

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Inspirational Example : GenAI AppStudio

Create GenAI Apps (Fully Functional LLM Powered Apps built using GenAI)

Power of GenAI

- Use GenAI to convert “sketches” to fully functional Apps
- Apps with an integrated RAG

Demo

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Building blocks of LLMs

Why Large?

- Large Training Dataset: Trained on massive datasets
- Large Architectures : Billions of parameters
- Large Computing Power: Requires massive GPUs

01 Foundation

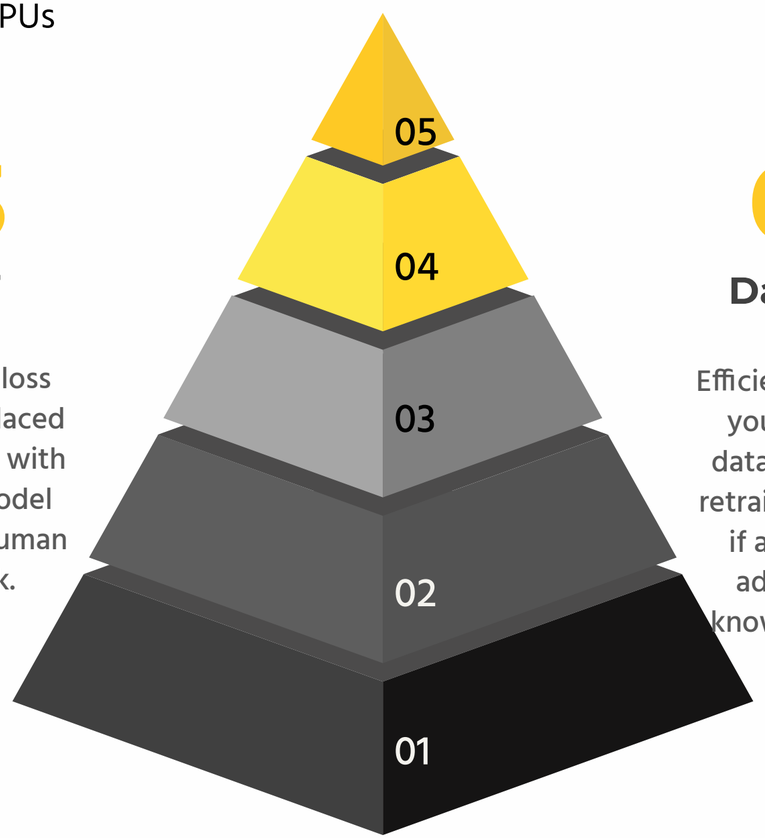
Enormous amount of text data trained in an autoregressive manner

02 Fine-tuning

Supervised fine-tuning on appropriate and well curated datasets to teach desired output behaviour.

03 RLHF

Next token loss function replaced or combined with a reward model trained on Human Feedback.



04 Database

Efficiently leverage your company data. No need to retrain your model if a new pdf is added to the knowledge base.

05 Memory

LLMs can have a huge context length and keep previous questions/tasks in memory for superior context understanding.

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Lab #1: Handson RAG

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Lab #2:
Data Prep for LLMs

Notebook - Data Prep, Aquarium Login, DataStudio, Doc 2 QA project

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10 Min Break

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Training Logistics

https://drive.google.com/file/d/1EngrX5JhNsejR6-e9KfdKiFtR1P8qlPU/view

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Lab #3: Fine Tuning

- Basic understanding of Fine-tuning
- Different hyperparameters of fine-tuning

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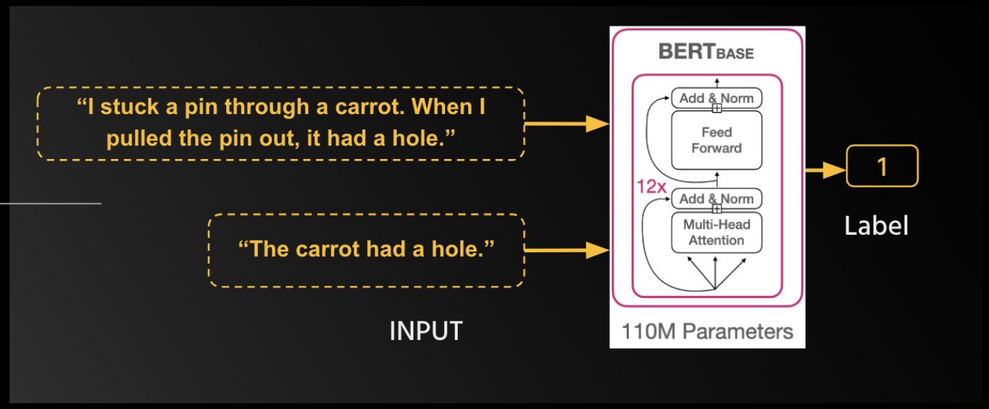
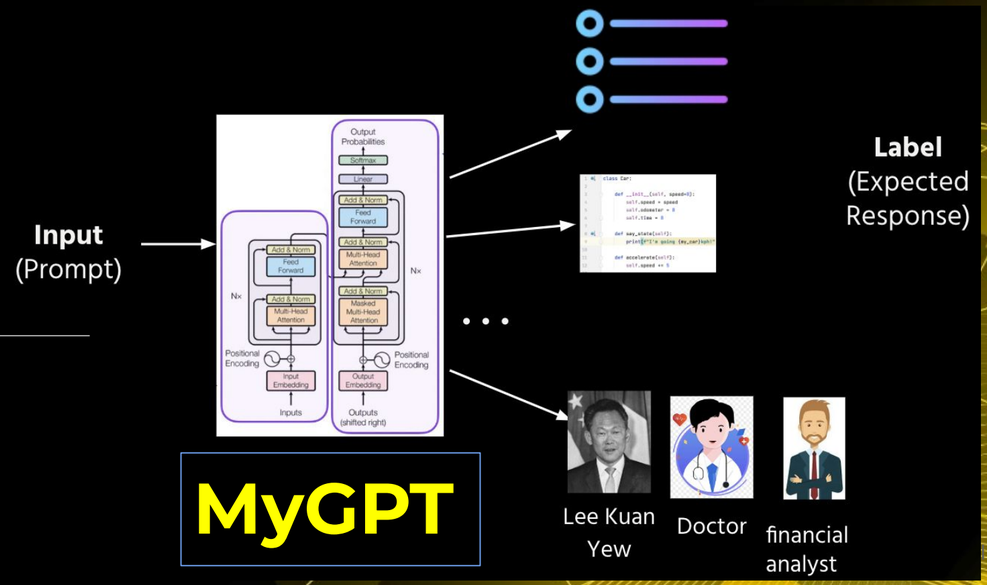
Fine Tuning of Language Models

Fine-tuning is the act of taking a pre-trained foundational model and training it on new data for a specific task.

Example tasks could be overarching NLP tasks like Text Summarization, classification etc.

and Instruct Tuning, as well as to induce a specific style in generated responses (e.g., talk like a sales agent, or social media influencer)





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LLMOps - Deployment and Ops

Customization | Inference | Resource Controls | Integration Flexibility | Compliance and Regulation

Build

Fine tune your LLM with **H2O LLM Studio**, and push it onto a **Model Hub** (**Hugging Face** or private).

Configure any prompt template, quantization method, etc. you want to include for your model.

Deploy

Register your model and version in H2O MLOps, select the **vLLM runtime**, and deploy!

Easily manage hardware resources according to your needs and usage.

Interact

Interact with the deployed model via single completion or chat-style conversations (compatible with **OpenAI API protocols**).

Integrate with popular toolkits, including **LangChain** and **Guardrails**.

Fine-tune your own LLM and let MLOps handle the serving!

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LLM Security is new and evolving

Jailbreaking
Prompt injection
Backdoors & data poisoning
Adversarial inputs
Insecure output handling
Data extraction & privacy
Data reconstruction
Denial of service
Watermarking & evasion
Model theft
...

Effective and continous output monitoring is Key!

Lab #4:
Making a GenAI App

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Lab #5:
Evaluate LLMs

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Quiz Link

https://forms.gle/96MuWcJJHtoNoEM76

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THANK YOU!